

45.2 Quantile regression

Least squares answers a question about a loss function: which constant minimizes expected squared error? Change the loss and the answer changes. The median minimizes expected absolute error, and one modification of absolute error — weighting the two sides differently — picks out any quantile at all. Koenker and Bassett (1978) built a regression method on that observation.

45.2.1 The check loss

Definition 45.2.1 — Check loss and quantile regression

For $\tau \in (0, 1)$ the **check function** is

$$\rho_\tau(u) = u\{\tau - \mathbf{1}\{u < 0\}\} = \begin{cases} \tau u, & u \geq 0, \\ (\tau - 1)u, & u < 0, \end{cases} \quad (45.2.1)$$

a convex, piecewise linear, nonnegative function with a kink at the origin, equal to $|u|/2$ when $\tau = 1/2$. The **linear quantile regression model** at level τ asserts

$$Q_Y(\tau \mid \mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}_\tau,$$

and given data $(y_i, \mathbf{x}_{(i)})$, $i = 1, \dots, n$, a **τ -regression quantile** is any minimizer

$$\hat{\boldsymbol{\beta}}_\tau \in \arg \min_{\mathbf{b} \in \mathbb{R}^p} R_\tau(\mathbf{b}) = \sum_{i=1}^n \rho_\tau(y_i - \mathbf{x}_{(i)}^\top \mathbf{b}). \quad (45.2.2)$$

The subscript τ is there because the coefficients may differ from level to level: the model is a separate statement for each τ , not one model with a nuisance parameter. Nothing is assumed about the shape of the conditional distribution, not even that it has a density.

45.2.2 What the check loss estimates

Theorem 45.2.1 — The check loss elicits the quantile

Let Y be a random variable with distribution function F and $E|Y| < \infty$, and let $G(q) = E \rho_\tau(Y - q)$.

(a) **Population characterization.** G is finite and convex, and for $q < q'$

$$G(q') - G(q) = \int_q^{q'} \{F(s) - \tau\} ds. \quad (45.2.3)$$

Consequently the set of minimizers of G is exactly the set of τ -quantiles of F , that is $\{q : F(q^-) \leq \tau \leq F(q)\}$.

- (b) **Linear program.** Writing $u_i = (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})_+$ and $v_i = (\mathbf{x}_{(i)}^\top \mathbf{b} - y_i)_+$, problem Eq. 45.2.2 is equivalent to

$$\min_{\mathbf{b}, \mathbf{u}, \mathbf{v}} \{ \tau \mathbf{1}^\top \mathbf{u} + (1-\tau) \mathbf{1}^\top \mathbf{v} \} \quad \text{subject to} \quad \mathbf{X}\mathbf{b} + \mathbf{u} - \mathbf{v} = \mathbf{y}, \quad \mathbf{u} \geq \mathbf{0}, \mathbf{v} \geq \mathbf{0}, \quad (45.2.4)$$

a linear program in $2n + 2p$ nonnegative variables with n equality constraints.

- (c) **Equivariance of the estimate.** Let $\hat{\beta}_\tau(\mathbf{y}, \mathbf{X})$ denote any solution of Eq. 45.2.2. For $a > 0$, $\boldsymbol{\gamma} \in \mathbb{R}^p$ and nonsingular $\mathbf{A}$,

$$\begin{aligned} \hat{\beta}_\tau(a\mathbf{y}, \mathbf{X}) &= a \hat{\beta}_\tau(\mathbf{y}, \mathbf{X}), & \hat{\beta}_\tau(-\mathbf{y}, \mathbf{X}) &= -\hat{\beta}_{1-\tau}(\mathbf{y}, \mathbf{X}), \\ \hat{\beta}_\tau(\mathbf{y} + \mathbf{X}\boldsymbol{\gamma}, \mathbf{X}) &= \hat{\beta}_\tau(\mathbf{y}, \mathbf{X}) + \boldsymbol{\gamma}, \\ \hat{\beta}_\tau(\mathbf{y}, \mathbf{X}\mathbf{A}) &= \mathbf{A}^{-1} \hat{\beta}_\tau(\mathbf{y}, \mathbf{X}). \end{aligned}$$

- (d) **Equivariance under monotone maps.** For strictly increasing left-continuous h , $Q_{h(Y)}(\tau | \mathbf{x}) = h\{Q_Y(\tau | \mathbf{x})\}$.

Proof.

- (a) Split the loss at the kink: $\rho_\tau(u) = \tau u_+ + (1-\tau)u_-$, where $u_- = (-u)_+$, so $G(q) = \tau E(Y - q)_+ + (1-\tau)E(q - Y)_+$, finite because both terms are bounded by $E|Y| + |q|$. By Fubini's theorem,

$$E(Y - q)_+ = \int_q^\infty \{1 - F(s)\} ds, \quad E(q - Y)_+ = \int_{-\infty}^q F(s) ds.$$

Subtracting the two expressions for q' and q gives

$$\begin{aligned} G(q') - G(q) &= -\tau \int_q^{q'} \{1 - F(s)\} ds + (1-\tau) \int_q^{q'} F(s) ds \\ &= \int_q^{q'} \{F(s) - \tau\} ds, \end{aligned}$$

which is Eq. 45.2.3; convexity follows because the integrand is nondecreasing. Let $q_\tau = \inf\{q : F(q) \geq \tau\}$. If $q < q_\tau$ then $F(s) < \tau$ for every $s < q_\tau$, so $G(q) - G(q_\tau) = \int_q^{q_\tau} \{\tau - F(s)\} ds > 0$ and q is not a minimizer. If $q > q_\tau$ then $F(s) \geq F(q_\tau) \geq \tau$ for $s \geq q_\tau$ by right continuity, so $G(q) \geq G(q_\tau)$, with equality if and only if $F \equiv \tau$ almost everywhere on (q_τ, q) , that is $F(q^-) = \tau$. Hence q_τ is a minimizer and the minimizers are exactly the points with $F(q^-) \leq \tau \leq F(q)$.

- (b) For any $\mathbf{b}$ the choices $u_i = (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})_+$ and $v_i = (\mathbf{x}_{(i)}^\top \mathbf{b} - y_i)_+$ are feasible with objective value $R_\tau(\mathbf{b})$, so the linear program's optimum is at most $\min_{\mathbf{b}} R_\tau(\mathbf{b})$. Conversely, if $(\mathbf{b}, \mathbf{u}, \mathbf{v})$ is feasible then $u_i - v_i = y_i - \mathbf{x}_{(i)}^\top \mathbf{b}$, and for a fixed difference the sum $\tau u_i + (1-\tau)v_i$ is minimized by $\min(u_i, v_i) = 0$, which gives exactly $\rho_\tau(y_i - \mathbf{x}_{(i)}^\top \mathbf{b})$. The two problems therefore have the same value and the same optimal $\mathbf{b}$.
- (c) For $a > 0$, $\rho_\tau(au) = a\rho_\tau(u)$, so $R_\tau(\mathbf{b}; a\mathbf{y}) = aR_\tau(\mathbf{b}/a; \mathbf{y})$ and the minimizers correspond under $\mathbf{b} \mapsto a\mathbf{b}$. A direct check on Eq. 45.2.1 gives $\rho_\tau(-u) = \rho_{1-\tau}(u)$, whence $R_\tau(\mathbf{b}; -\mathbf{y}) = R_{1-\tau}(-\mathbf{b}; \mathbf{y})$ and the second identity. For the shift, $y_i + \mathbf{x}_{(i)}^\top \boldsymbol{\gamma} - \mathbf{x}_{(i)}^\top \mathbf{b} = y_i - \mathbf{x}_{(i)}^\top (\mathbf{b} - \boldsymbol{\gamma})$, so the objective is the old one at $\mathbf{b} - \boldsymbol{\gamma}$. For the reparameterization, the residuals of $\mathbf{X}\mathbf{A}$ at $\mathbf{b}$ are those of $\mathbf{X}$ at $\mathbf{A}\mathbf{b}$.

(d) This is Lemma 45.1.1 applied conditionally on $\mathbf{x}$.

■

Part (a) is the whole justification of the method, and it needs remarkably little: no density, no moments beyond the first, no continuity. Part (d) is a property least squares does not have: the median of $\log Y$ is the logarithm of the median of Y , while the mean of $\log Y$ is not the logarithm of the mean, the retransformation problem of [Section 22.5](#). A quantile regression fitted on the log scale reads back without a smearing correction.

Remark (The influence of one observation). Away from the kink $\rho'_\tau(u) = \tau - \mathbf{1}\{u < 0\}$ takes only the values τ and $\tau - 1$, so an observation enters the estimating equation *only through the sign of its residual*: moving one response from $+10$ to $+10^6$ changes nothing. This is the bounded-influence property that made the median robust in [Section 20.6](#). It buys nothing against a bad *design* point, whose $\mathbf{x}_{(i)}$ enters unbounded.

Remark (Without moments). Minimizing $G_0(q) = E\{\rho_\tau(Y - q) - \rho_\tau(Y)\}$ instead removes the moment condition: since $|\rho_\tau(u - v) - \rho_\tau(u)| \leq \max(\tau, 1 - \tau)|v|$, this is finite whatever the tails of Y , and differs from G by a constant whenever G is finite, so (a) holds verbatim.

45.2.3 Solving the problem

Proposition 45.2.2 — Interpolation and counting

Let $\text{rank}(\mathbf{X}) = p$ and let $\hat{\boldsymbol{\beta}}_\tau$ minimize [Eq. 45.2.2](#).

- (a) There is a minimizer whose residual vector vanishes at p observations whose rows of $\mathbf{X}$ are linearly independent: the fit passes exactly through p of the data points.
- (b) If $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, then for *any* minimizer, with n_-, n_0, n_+ the numbers of negative, zero and positive residuals,

$$n_- \leq \tau n \leq n_- + n_0. \quad (45.2.5)$$

Proof.

- (a) Let $\mathbf{b}$ be any minimizer, $H = \{i : y_i = \mathbf{x}_{(i)}^\top \mathbf{b}\}$ the interpolated set, and suppose $\{\mathbf{x}_{(i)} : i \in H\}$ spans a space of dimension less than p . Choose $\mathbf{d} \neq \mathbf{0}$ orthogonal to all of them. For small t the residuals in H stay zero and no other residual changes sign, so

$$R_\tau(\mathbf{b} + t\mathbf{d}) = \sum_{i \notin H} (r_i - t\mathbf{x}_{(i)}^\top \mathbf{d}) \{\tau - \mathbf{1}\{r_i < 0\}\}, \quad r_i = y_i - \mathbf{x}_{(i)}^\top \mathbf{b},$$

which is affine in t . An affine function minimized at $t = 0$ is constant, so R_τ stays minimal as t grows, until some residual outside H reaches zero. Because $\text{rank}(\mathbf{X}) = p$ and $\mathbf{d} \neq \mathbf{0}$, some $\mathbf{x}_{(i)}^\top \mathbf{d} \neq 0$, so such a t exists and is finite; there the interpolated set is strictly larger and the objective still minimal. Repeating raises the rank of $\{\mathbf{x}_{(i)} : i \in H\}$ to p .

- (b) Let $\mathbf{1} = \mathbf{X}\mathbf{c}$ and consider $\phi(t) = R_\tau(\hat{\boldsymbol{\beta}}_\tau + t\mathbf{c})$, which shifts every fitted value

by t and so every residual by $-t$. For $t > 0$ small, an observation with $r_i > 0$ contributes $\tau(r_i - t)$, one with $r_i \leq 0$ contributes $(\tau - 1)(r_i - t)$; hence $\phi'(0^+) = -\tau n_+ + (1 - \tau)(n_- + n_0)$. Similarly $\phi'(0^-) = -\tau(n_+ + n_0) + (1 - \tau)n_-$. Minimality forces $\phi'(0^+) \geq 0$ and $\phi'(0^-) \leq 0$. The first gives $\tau n_+ \leq (1 - \tau)(n_- + n_0)$; adding $\tau(n_- + n_0)$ to both sides and using $n_+ + n_- + n_0 = n$ yields $\tau n \leq n_- + n_0$. The second gives $\tau(n_+ + n_0) \geq (1 - \tau)n_-$, and adding τn_- to both sides yields $\tau n \geq n_-$. ■

Part (b) says the fitted surface has about τn observations below it, which is what the name promises; part (a) says it is pinned to p of the data points, as the median of an odd number of points is one of the points. Together they make the fit a combinatorial object: finitely many candidate hyperplanes, one per choice of p rows, and the optimum is one of them. Hence the simplex method of Koenker and d'Orey (1987) rather than differentiation, or the interior-point route of Portnoy and Koenker (1997), which makes quantile regression competitive in cost with least squares for large n .

Example. (Seven quantile lines for Engel's budgets)

Fitting Eq. 45.2.4 at seven levels gives slopes rising steadily with τ :

τ	0.05	0.10	0.25	0.50	0.75	0.90	0.95
slope	0.3434	0.4018	0.4741	0.5602	0.6440	0.6863	0.7091
intercept	124.9	110.1	95.5	81.5	62.4	67.4	64.1

The least squares slope, 0.4852, sits between the quartiles. A household at the tenth percentile of its income group spends about 0.4018 of each extra franc on food, one at the ninetieth 0.6863; at an income of 1000 francs the fitted median budget share is 0.642 and the tenth-percentile share 0.512. Every fit interpolates 2 observations and satisfies Eq. 45.2.5, as the listings verify.

```
import numpy as np
import statsmodels.api as sm
from scipy.optimize import linprog
data = sm.datasets.engel.load_pandas().data
TAUS = (0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95)

income = data["income"].to_numpy()
food = data["foodexp"].to_numpy()
n = len(food)
X = np.column_stack([np.ones(n), income])

def check_loss(u, tau):
    """The check function rho_tau applied elementwise."""
    return u * (tau - (u < 0))

def qreg(X, y, tau):
    """Quantile regression by linear programming.

    Variables are (b+, b-, u, v), all nonnegative, with beta = b+ - b- and
    u - v = y - X beta the positive and negative parts of the residual.
    """
```

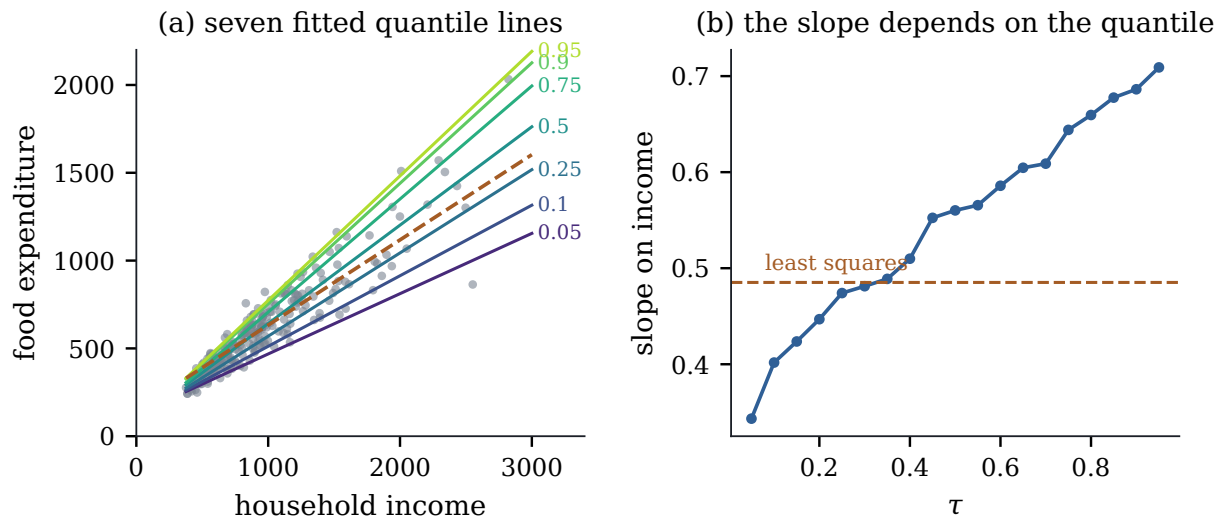


Figure 45.2.1: (a) Seven fitted quantile lines for Engel's budgets, least squares dashed. (b) The fitted slope against τ .

```
n, p = X.shape
cost = np.concatenate([np.zeros(2 * p), tau * np.ones(n), (1 - tau) * np.ones(n)
])
A = np.hstack([X, -X, np.eye(n), -np.eye(n)])
out = linprog(cost, A_eq=A, b_eq=y, bounds=(0, None), method="highs")
return out.x[:p] - out.x[p:2 * p]

beta = {tau: qreg(X, food, tau) for tau in TAUS}
for tau in (0.10, 0.50, 0.90):
    print(f"tau = {tau:.2f}:  intercept {beta[tau][0]:8.2f}    slope {beta[tau][1]:.4f}")
```

[Run this code online](#)

```
for tau in TAUS:
    r = food - X @ beta[tau]
    n_zero = np.sum(np.abs(r) < 1e-8)
    n_neg = np.sum(r < -1e-8)
    assert n_zero == X.shape[1] # p residuals vanish exactly
    assert n_neg <= tau * n <= n_neg + n_zero # the counting identity
    print("at every tau: 2 residuals are zero and #negative <= tau*n <= #negative + 2")
```

[Run this code online](#)

Koenker and Machado (1999) proposed the analogue of R^2 : with $\hat{V}(\tau)$ the minimized check loss and $\tilde{V}(\tau)$ its value in the intercept-only model, $R_1(\tau) = 1 - \hat{V}(\tau)/\tilde{V}(\tau)$. For Engel's data it rises from 0.474 at $\tau = 0.05$ to 0.805 at $\tau = 0.95$: income explains far more about the upper tail than about the lower. Like R^2 it says nothing about whether the linear specification is right.

45.2.4 Reading a coefficient

The coefficient $\beta_{\tau j}$ is the derivative of the τ -quantile of the conditional distribution with respect to x_j . It is *not* the effect of x_j on the household that happens to sit at rank τ .

Remark (Quantiles of distributions, not of people). Suppose incomes rise by one franc for every household. The 0.9-quantile of the new conditional distribution exceeds the old by $\beta_{0.9}$; but the household at rank 0.9 before need not be there after, and nothing in the data says what happened to it. Reading β_τ as an individual effect assumes the change preserves ranks — the kind of untestable structure [Chapter 25](#) insists on making explicit (Definition 25.2.1). Without it, quantile regression describes how a *distribution* moves, which is usually the question anyway.

Two cautions carry over: extrapolation is as unsafe as in [Section 12.3](#), and [Section 45.3](#) shows it is the first place quantile curves misbehave; and linearity of $Q_Y(\tau \mid \mathbf{x})$ is an assumption, testable in the spirit of Theorem 14.6.2.

45.2.5 Exercises

45.2.5.1 A. Check your understanding

Exercise A1.

A quantile regression with an intercept and one regressor is fitted at $\tau = 0.25$ to $n = 40$ observations. How many residuals can be negative, and how many are exactly zero? Use [Proposition 45.2.2](#).

Solution. Generic data give $n_0 = p = 2$, and [Eq. 45.2.5](#) reads $n_- \leq 10 \leq n_- + 2$, so $n_- \in \{8, 9, 10\}$.

45.2.5.2 B. Practice

Exercise B1.

Take $\mathbf{X} = \mathbf{1}$, so that the problem is to fit one number. Show that the minimizers of $\sum_i \rho_\tau(y_i - b)$ are exactly the τ -quantiles of the empirical distribution of $y_1, \dots, y_n$, and that for τn not an integer the minimizer is the unique order statistic $y_{(\lceil \tau n \rceil)}$.

Solution. Apply [Theorem 45.2.1\(a\)](#) to the empirical distribution F_n , which is legitimate because F_n is a distribution function with finite mean. The minimizers are $\{b : F_n(b^-) \leq \tau \leq F_n(b)\}$. If $\tau n \notin \mathbb{Z}$, no order statistic has F_n equal to τ , and the condition picks out the single point $y_{(\lceil \tau n \rceil)}$; if τn is an integer, the whole interval $[y_{(\tau n)}, y_{(\tau n+1)}]$ minimizes.

Exercise B2.

Write down the dual of the linear program [Eq. 45.2.4](#) and show that it is

$$\max_{\mathbf{d}} \mathbf{y}^\top \mathbf{d} \quad \text{subject to} \quad \mathbf{X}^\top \mathbf{d} = \mathbf{0}, \quad \mathbf{d} \in [\tau - 1, \tau]^n,$$

or, after the shift $\mathbf{a} = \mathbf{d} + (1 - \tau)\mathbf{1}$, $\max\{\mathbf{y}^\top \mathbf{a} : \mathbf{X}^\top \mathbf{a} = (1 - \tau)\mathbf{X}^\top \mathbf{1}, \mathbf{a} \in [0, 1]^n\}$. Show that at an optimum $a_i = 1$ for a positive residual and $a_i = 0$ for a negative one, so that

a records the “signs” of the residuals, and explain why these are the raw material of the rank tests of [Section 45.3](#).

45.2.5.3 C. Going deeper

Exercise C1.

Show that the set of minimizers of R_τ is a convex polyhedron, and that if it contains two distinct points then R_τ is constant on the segment joining them. Deduce that a unique solution is the generic case: non-uniqueness requires a coincidence in the data, such as $p + 1$ observations lying on one hyperplane.